Inequalities and equality cases

High school · optimisation and olympiads · Worked solutions · 11

Exercise 1. For $x > 0$, prove $x + \frac{1}{x} \geq 2$ and determine equality.

Solution. $x + \frac{1}{x} - 2 = \frac{(x-1)^2}{x} \geq 0$. Equality holds exactly at $x = 1$.

Exercise 2. Find the minimum of $x^2 - 6x + 13$ on $\mathbb{R}$.

Solution. $x^2 - 6x + 13 = (x-3)^2 + 4 \geq 4$. The minimum 4 is attained at $x = 3$.

Exercise 3. A rectangle has perimeter 24 m. What is its greatest area?

Solution. Its sides $a, b > 0$ satisfy $a + b = 12$. Thus $ab \leq 36$, with equality at $a = b = 6\,\text{m}$. The maximum area is $36\,\text{m}^2$.

Exercise 4. For real a,b, prove $a^2 + b^2 \geq 2ab$. Must a and b be positive?

Solution. $a^2 + b^2 - 2ab = (a-b)^2 \geq 0$ for all real a,b. No positivity is needed; equality means $a = b$.

Problem · Investigate, justify, conclude. There are 40 m of fencing for three sides of a rectangular enclosure against a wall. Its width x, perpendicular to the wall, is positive. Express its length and area in terms of x, determine the domain, then prove the maximum area without derivatives. Compare with a fully fenced rectangle of perimeter 40 m.

Solution. Length is $40 - 2x$ and $0 < x < 20$. Area is $A(x) = x(40 - 2x) = 200 - 2(x-10)^2 \leq 200$. Equality occurs at width 10 m and length 20 m: $200\,\text{m}^2$. With all four sides fenced, $a + b = 20$ and the maximum area is $100\,\text{m}^2$, attained by a square of side 10 m.